

Digital Forensics Laboratory

EXPERIMENT NO: 04

DATE: August 6, 2026

TITLE: Analyze Email Headers and Detect Email Spoofing using MHA (Mail Header Analyzer)

1. AIM

To extract, analyze, and interpret email headers to trace an email's origin, verify its authenticity protocols (SPF, DKIM, DMARC), and detect potential spoofing or phishing attempts using Mail Header Analyzer (MHA) tools.

2. SOFTWARE / TOOLS REQUIRED

Operating System: Windows, Linux, or macOS

Email Client: Web browser accessing Gmail, Outlook, or Yahoo

Analysis Tools: MXToolbox (Online Mail Header Analyzer), WHOIS IP Lookup

3. THEORY

Every email sent over the internet contains hidden metadata known as an **Email Header**. This header tracks the email's journey from the sender's server to the recipient's inbox and contains critical routing and authentication information. Analyzing these headers helps forensic investigators verify if an email is legitimate or spoofed.

Key Authentication Protocols:

SPF (Sender Policy Framework): Checks if the sender's IP address is authorized to send emails on behalf of the claimed domain.

DKIM (DomainKeys Identified Mail): A cryptographic signature that verifies the email content has not been altered in transit.

DMARC (Domain-based Message Authentication, Reporting, and Conformance): A policy that uses both SPF and DKIM to authenticate the email and instruct the receiving server on how to handle failures.

Figure 2: Extracting the original raw message and email header data

Step 3: Analyzing the Header via MXToolbox

1. Navigate to an online Mail Header Analyzer, such as **MXToolbox**.
2. Paste the copied raw header text into the analysis tool and click Analyze.
3. Review the **Delivery Information** and **Relay Information** to trace the chronological path (hops) the email took from the sending server to the receiving server. Check for unusual delays or suspicious routing.



Figure 3: MHA relay analysis showing chronological hops and delivery delays

Step 4: Examining SPF, DKIM, and DMARC Results

1. Scroll down in the analyzer tool to inspect the authentication policies and Verify if the DMARC record is compliant.
2. Check the SPF and DKIM alignment to ensure the sender's IP (**103.52.180.165**) was genuinely authorized by the domain (communication.vodafoneidea.in).

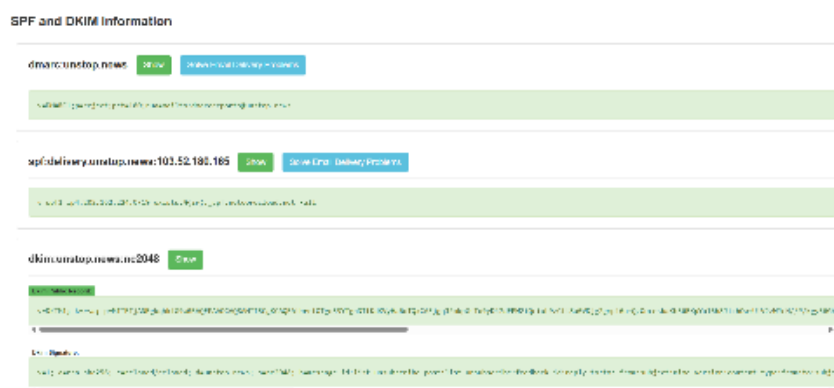


Figure 4: Verification of SPF, DKIM, and DMARC cryptographic records

Step 5: IP Address and Hostname Verification

1. Extract the primary sending IP address identified in the Received lines (103.52.180.165).
2. Use a **WHOIS** lookup tool to identify the geographical location and ownership of the IP address.
3. Determine if the registered organization matches the expected sender.

Whois IP 103.52.180.165

Updated 1 second ago

```
% [whois.apnic.net]
% whois data copyright Terms      http://www.apnic.net/db/dbcopyright.html

% Information related to '103.52.180.0 - 103.52.183.255'

% Abuse contacts for '103.52.180.0 - 103.52.183.255' are 'abuse@mumbai.ravience.in', 'abuse@netcore'

inetnum:        103.52.180.0 - 103.52.183.255
netname:        NETCORE-IN
descr:          Netcore Solutions Pvt Ltd
descr:          8th Floor, Peninsula Tower, Peninsula Corporate Park, G.K. Marg, Lower Parel (W), Mui
admin-c:        MK478-AP
tech-c:         KJ35-AP
country:        IN
mnt-by:         MAINT-IN-IRINN
mnt-lower:      MAINT-IN-NETCORE-INC
mnt-lower:      MAINT-IN-IRINN
mnt-routes:     MAINT-IN-NETCORE-INC
mnt-irt:        IRT-NETCORE-IN
status:         ALLOCATED PORTABLE
last-modified:  2025-08-11T22:49:15Z
source:         APNIC

irt:            IRT-NETCORE-IN
address:        402, PENINSULA CHAMBERS,
address:        G.K. MARG,
address:        LOWER PAREL (W),
address:        MUMBAI - 400013
e-mail:         admin@mumbai.ravience.in
e-mail:         kuldeep@netcorecloud.com
abuse-mailbox:  abuse@mumbai.ravience.in
abuse-mailbox:  abuse@netcorecloud.com
```

Figure 5: WHOIS IP lookup revealing the hosting organization (Netcore-In)

5. FORENSIC FINDINGS & ANALYSIS

Based on the header analysis conducted:

Sender Domain: communication.vodafoneidea.in

Sending IP Address: 103.52.180.165

Authentication Status: The email **PASSED** all primary checks (SPF, DKIM, and DMARC). The IP address is officially authorized to send on behalf of the

domain.

WHOIS Verification: The IP belongs to Netcore-In, Netcore is the email-delivery infrastructure, not necessarily the actual sender of the job advertisement. In this case, the apparent sender is Team Unstop.

Conclusion on Spoofing: Despite being flagged as spam by Gmail's automated content filters (likely due to promotional marketing content), the header analysis proves this email is **cryptographically authentic and not spoofed**. The sender identity is genuine.

6. RESULT

The email header was successfully extracted and analyzed using MXToolbox. Authentication protocols (SPF, DKIM, DMARC) and the originating IP address were

manually verified, demonstrating how to trace email origins and detect potential spoofing.